

Mechanism Design via Market-Clearing Prices for Value Maximizers under Budget and RoS Constraints

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Abstract. The growing adoption of auto-bidding in online advertising has shifted the focus of auction theory from quasi-linear utility maximization to value maximization subject to financial constraints. Advertisers increasingly seek to maximize the total value while respecting to financial constraints. These new transition raises new mechanism questions.

In this paper, we study a setting where buyers have private budgets and private return over spend (RoS) constraints, while valuations are public and drawn from a continuous distribution. This information structure is motivated by modern advertising platforms, where machine learning models predict auction valuations, while advertisers set budget and RoS constraints that are not known to the platform.

We first extend the Eisenberg-Gale convex program to incorporate RoS constraints and show that its unique solution characterizes a competitive equilibrium with common item-level prices. Building on this characterization, we construct a market-clearing mechanism under which truthful reporting of both budget and RoS is ex-post optimal. The mechanism achieves a tight $\frac{1}{2}$ -approximation to the first-best revenue benchmark, defined as the maximum revenue attainable under buyers' true budget and RoS constraints without imposing incentive compatibility. Thus, our mechanism provides a revenue guarantee despite the privacy of buyers' financial constraints.

We further show that the market-clearing mechanism's allocations and prices coincide with the equilibrium outcome of a first-price auction under uniform bidding. This equivalence provides a simple implementation of the market clearing mechanism.

Finally, we study how the market clearing mechanism can be implemented in an online setting. We design a decentralized algorithm that learns the equilibrium multiplier using only bandit feedback. Over m auctions, both the seller's revenue regret and each buyer's total value regret are $\tilde{O}(\sqrt{m})$, comparing to the equilibrium revenue and total value, respectively. Matching $\Omega(\sqrt{m})$ lower bounds establish that these rates are optimal up to logarithmic factors. The full version is available at <https://arxiv.org/abs/2602.19085>

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